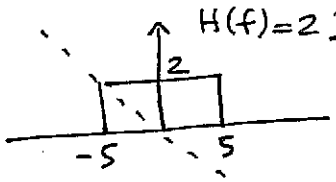
c) 

$H(f) = 2\Lambda\left(\frac{f}{10}\right) e^{-j2\pi f.4}$

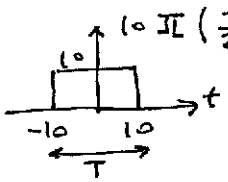
It is not ideal because $|H(f)| \neq |K|$ $\forall f$

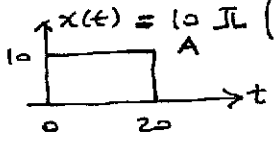
NAME: KEY NUMBER:

**ELEC361 Analog Communication Systems Midterm Exam** **Fall 2019**

**Prof. Oğuz Kucur** **Time: 100 minutes** **18.11.2019**

**1)(30 p.) a)** For the signal $x(t) = 10\prod\left(\frac{t-10}{20}\right)$, find autocorrelation function, energy spectral density (ESD) function and average energy. (Sketch $x(t)$ and its ESD function.) **b)** If this signal passes through a linear time invariant (LTI) system with the impulse response $h(t) = 20Sinc(10t-40)$, find the ESD function of the output signal $y(t)$. Find the average power of $y(t)$. (Sketch the ESD function.) **c)** Sketch the spectra of the LTI system. If this system is a channel, is it ideal (distortionless) or not? Why?

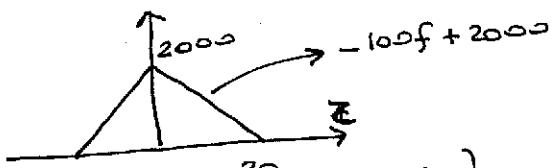
a) 

$\Rightarrow$ 


$X(f) = AT\,Sinc(fT)e^{-j2\pi f t_0} = 10.20\,Sinc(20f)\,e^{-j2\pi f.10}$

ESD: $\mathcal{E}_x(f) = |X(f)|^2 = \underbrace{40000}_{A.T=20}\,Sinc^2(20f) \xrightarrow{\mathcal{F}^{-1}} R_x(\tau) = A\Lambda\left(\frac{t}{T}\right) = 2000\,\Lambda\left(\frac{t}{20}\right)$

$R_x(\tau)$




$E_x = R_x(0) = 2000\,J$

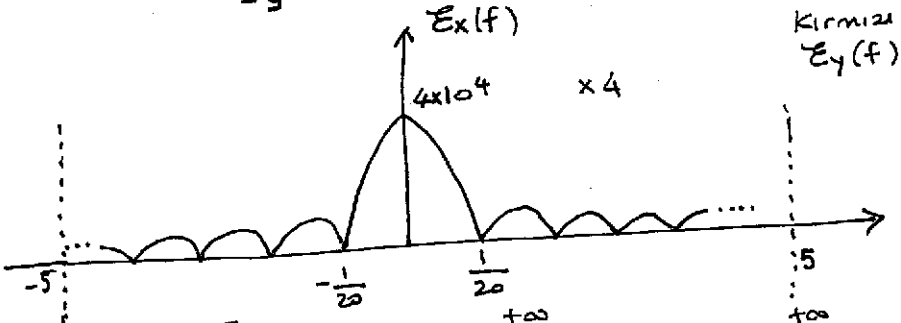
or $= \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_0^{20} 10^2 dt = 2000\,J$

(Sınavda çizmeniz istenmiyordu.)

b) $h(t) = \underbrace{20}_{A.T} Sinc\left[\underbrace{10}_{T}(t-\underbrace{4}_{t_0})\right] \xrightarrow{\mathcal{F}} H(f) = A\,\Pi\left(\frac{f}{T}\right)e^{-j2\pi f t_0} = 2\,\Pi\left(\frac{f}{10}\right)e^{-j2\pi f.4}$

$\mathcal{E}_y(f) = |H(f)|^2.\,\mathcal{E}_x(f) = \begin{cases} 4\,\mathcal{E}_x(f), & -5 \le f \le 5\ Hz \\ 0, & Diğer \end{cases}$

$P_y = \int_{-5}^{5} \mathcal{E}_y(f)df = \int_{-5}^{5} 4\times 40\,000\,Sinc^2(20f)df$




Kırmızı renk $\mathcal{E}_y(f)$ için.

Sinc(.) fonksiyonunun enerjisinin %90'ı ana lobundadır. $5 \gg \frac{1}{20}$ olduğu için integralin sınırlarını $\infty$ kabul edebiliriz.

$P_y = \int_{-5}^{5} \mathcal{E}_y(f)df \cong \int_{-\infty}^{+\infty} \mathcal{E}_y(f)df = \int_{-\infty}^{+\infty} 4\times 40\,000\,Sinc^2(20f)df = \int_{-\infty}^{\infty} 16\times 10^4\left[\frac{1}{20}\Pi\left(\frac{t}{20}\right)\right]^2 dt$

$\xrightarrow{}$ Parseval teo.

$= \frac{16\times 10^4}{400}\int_{-10}^{10} 1^2 dt = \frac{16\times 10^4}{400}.20 = 8000\,J$

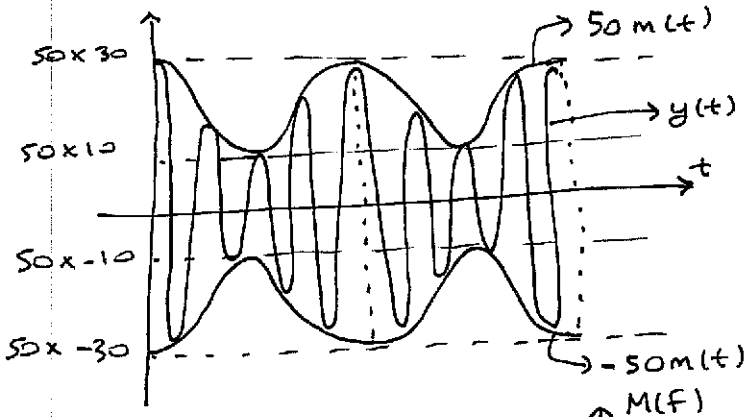
2)(30 p.) The message signal $m(t) = 20 + 10Cos(2000\pi t)$ modulates the carrier signal $c(t) = 50Cos(20000\pi t)$ to obtain DSB-SC-AM signal. a) Find and sketch the modulated signal $y(t)$. Find Fourier transforms of $m(t)$ and $y(t)$. Sketch their spectra. Find the power and bandwidth of $y(t)$. b) Find the signals at the input and output of the LPF in both time and frequency domains if the carrier signal is generated at the demodulator as $Cos(20000\pi t + \frac{\pi}{4})$. What is the bandwidth of the LPF? Find the power of the signal at the output of the LPF and comment on it.

$m(t) = 20 + 10\ Cos(2\pi\ 1000\ t) \rightarrow BW\{m(t)\} = B = 1\ kHz$

$c(t) = \underset{A_c}{50}\ Cos(2\pi\ \underset{f_c}{10000}\ t)$

a) $y(t) = m(t).\ c(t) = 50\ m(t)\ Cos(2\pi\ 10k\ t)$

$= 1000\ Cos(2\pi\ 10k\ t) + 250\ [Cos(2\pi\ 11k\ t) + Cos(2\pi\ 9k\ t)]$




$M(f) = 20\ \delta(f) + 5\ [\delta(f - 1k) + \delta(f + 1k)]$

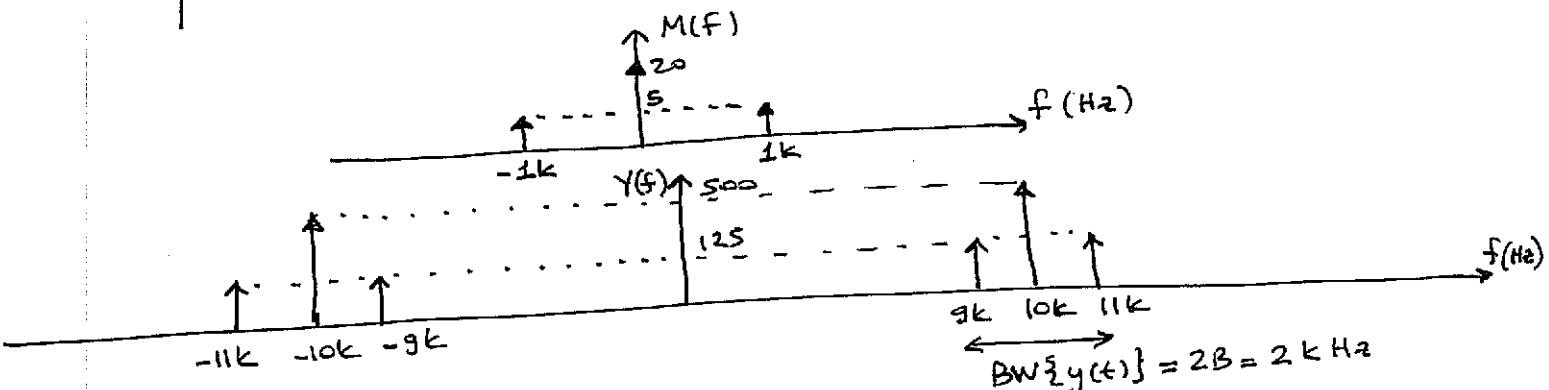
$Y(f) = 500\ [\delta(f - 10k) + \delta(f + 10k)] +$

$125\ [\delta(f - 11k) + \delta(f + 11k) + \delta(f - 9k) + \delta(f + 9k)]$

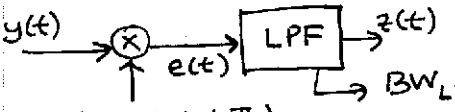
$P_y = \frac{A_c^2}{2} P_m$

$P_m = 20^2 + 10^2\ \frac{1}{2} = 450$ watt

$P_y = \frac{50^2}{2} \cdot 450 = 562,5$ kW




b)




$e(t) = y(t).\ Cos(2\pi\ 10k\ t + \frac{\pi}{4}) = 50\ m(t)\ Cos(2\pi\ 10k\ t).\ Cos(2\pi\ 10k\ t + \frac{\pi}{4})$

$= 25\ m(t)\ [Cos(\pi/4) + Cos(2\pi\ 20k\ t + \pi/4)] = 25 m(t) Cos(\pi/4) + 25 m(t) Cos(2\pi\ 20k\ t + \pi/4)$

Don't pass through LPF

$z(t) = 25\ m(t)\ Cos(\pi/4) = \frac{25\sqrt{2}}{2} m(t) \rightarrow Z(f) = 12,5\sqrt{2}\ M(f)$

$E(f) = 12,5\sqrt{2}\ M(f) + 12,5\ [M(f - 20k)\ e^{j\pi/4} + M(f + 20k)\ e^{-j\pi/4}]$

$P_z = (12,5\sqrt{2})^2\ P_m = 140,625$ kW

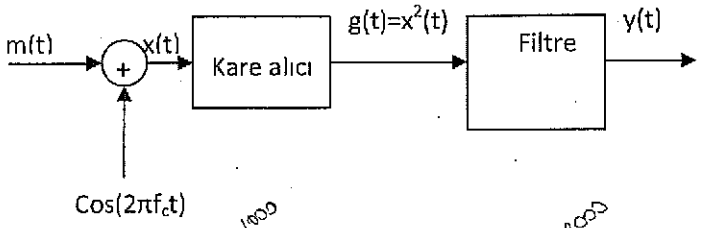
Comment: Transmitted power $P_y = 562,5$ kW, demodulated power $P_z = 140,625$ kW

Since the carrier is not synchronized at the receiver, the demodulated power decreased by $Cos^2(\pi/4) = 1/2$.

$\cos^2\alpha = \frac{1+\cos 2\alpha}{2}$ $\quad \cos a.\cos b = \frac{1}{2}\cos(a+b) + \frac{1}{2}\cos(a-b)$

a)20
b)10
c)10

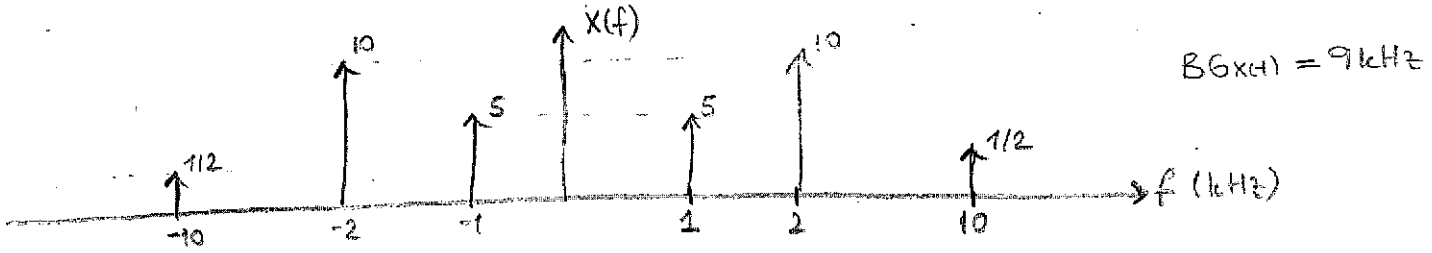
**4) (40 p.)** $m(t) = 10\,\mathrm{Cos}\,(2000\pi t) + 20\,\mathrm{Cos}(4000\pi t)$ olarak verilen bir mesaj işareti aşağıdaki şekilde görülen Taşıyıcısı Bastırılmış (TB)-ÇYB-GM modülatöründen geçmektedir. ($f_c = 10000\ Hz$) a) Filtre nasıl bir filtre olmalıdır? Özelliklerini belirtiniz. $x(t), g(t), y(t)$ işaretlerini zaman ve frekans domenlerinde ifade ediniz, spektrumlarını çiziniz ve band genişliklerini bulunuz. b) Demodülatör yapısını çiziniz ve her adımda elde edilen işareti zaman ve frekans domenlerinde ifade ediniz, spektrumlarını çiziniz. c) Alıcıda taşıyıcı işaret $Cos[2\pi(f_c + \Delta f)t + \theta]$ olarak üretilirse demodülatör çıkışındaki işareti zaman domeninde bulunuz. i) $\Delta f = 100\ Hz,\ \theta = 0$ için yorumlayınız. ii) $\Delta f = 0,\ \theta = \pi/2$ için yorumlayınız.




a) $X(t) = m(t) + \cos 2\pi f_c t = 10\cos(2000\pi t) + 20\cos(4000\pi t) + \cos(20000\pi t)$

$X(f) = 5\,\delta(f-1000) + 5\,\delta(f+1000) + 10\,\delta(f-2000) + 10\,\delta(f+2000)$

$+\frac{1}{2}\delta(f-10000) + \frac{1}{2}\delta(f+10000)$




$B_{G_{X(t)}} = 9\,kHz$

$g(t) = X^2(t) = m^2(t) + \cos^2(2\pi f_c t) + 2m(t)\cos(2\pi f_c t)$

$g(t) = 100\cos^2(2000\pi t) + 400\cos^2(4000\pi t) + 400\cos(2000\pi t)\cos(4000\pi t) + \cos^2(2\pi f_c t)$

$+ 20\cos(2000\pi t)\cos(2\pi f_c t) + 40\cos(4000\pi t)\cos(2\pi f_c t)$

$g(t) = 50\left[1+\cos(4000\pi t)\right] + 200\left[1+\cos(8000\pi t)\right] + \frac{1}{2}\left[1+\cos(2\pi\, 2f_c t)\right]$

$+ 200\cos(6000\pi t) + 200\cos(2000\pi t) + 10\cos(22000\pi t) + 10\cos(18000\pi t)$

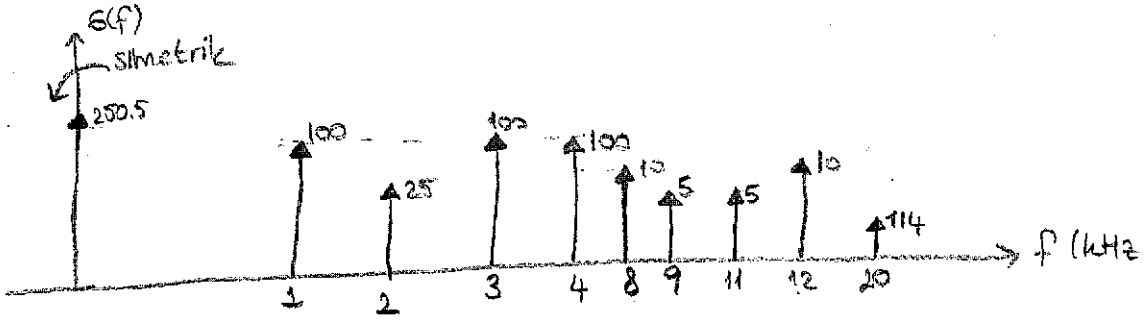
$+ 20\cos(24000\pi t) + 20\cos(16000\pi t)$

$$G(f) = 50\delta(f) + 25\delta(f-2000) + 25\delta(f+2000) + 200\delta(f) + 100\delta(f-4000) + 100\delta(f+4000)$$

$$+\frac{1}{2}\delta(f) + \frac{1}{4}\delta(f-20000) + \frac{1}{4}\delta(f+20000) + 100\delta(f-3000) + 100\delta(f+3000)$$

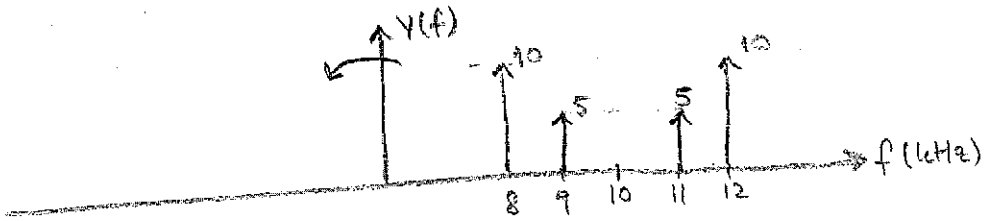
$$+100\delta(f-1000) + 100\delta(f+1000) + 5\delta(f-11000) + 5\delta(f+11000) + 5\delta(f-9000) + 5\delta(f+9000$$

$$+10\delta(f-12000) + 10\delta(f+12000) + 10\delta(f+8000) + 10\delta(f-8000)$$




$BG_{g(t)} = 19\ kHz$

Filtre → Merkez frekansı : 10 kHz, BG = 4 kHz } olan bir BGF olmalı



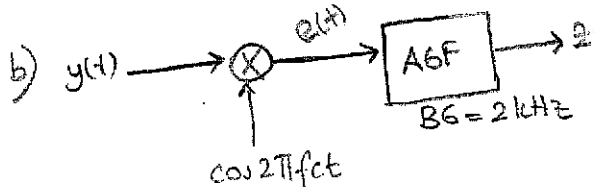

$BG_{y(t)} = 4\ kHz$

$$Y(f) = 5\delta(f-9) + 5\delta(f+9) + 5\delta(f-11) + 5\delta(f+11) + 10\delta(f-12) + 10\delta(f+12)$$

$$+10\delta(f-8) + 10\delta(f+8)$$

$$y(t) = 10\cos(18000\pi t) + 10\cos(22000\pi t) + 20\cos(24000\pi t) + 20\cos(16000\pi t)$$

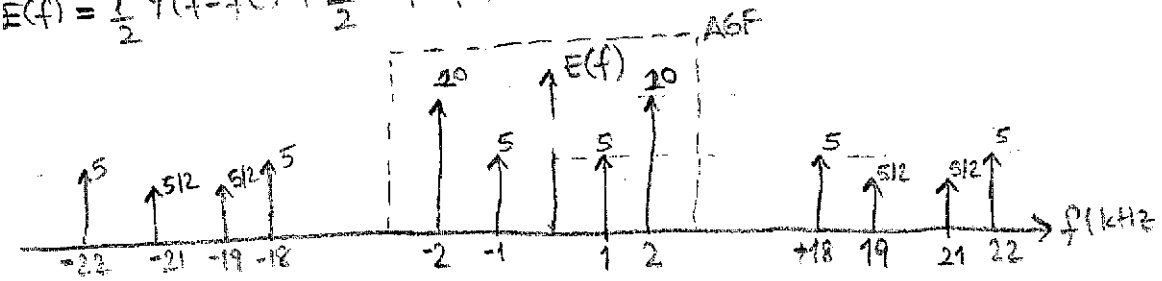
b)




$z(t) = 10\cos(2000\pi t) + 20\cos(4000\pi t)$

$e(t) = y(t)\cos 2\pi f_c t$

$E(f) = \frac{1}{2}Y(f-f_c) + \frac{1}{2}Y(f+f_c)$



3) $e(t) = y(t) \cdot \cos\left[2\pi(f_c + \Delta f)t + \theta\right]$

$y(t) = 10\cos(18000\pi t) + 10\cos(22000\pi t) + 20\cos(24000\pi t) + 20\cos(16000\pi t)$

9k 11k 12k 8k

$e(t) = 5\cos\left(2\pi(19000+\Delta f)t+\theta\right) + 5\cos\left(2\pi(1000+\Delta f)t+\theta\right)$

$+ 5\cos\left(2\pi(21000+\Delta f)t+\theta\right) + 5\cos\left(2\pi(1000+\Delta f)t+\theta\right)$

$+10\cos\left(2\pi(22000+\Delta f)t+\theta\right) + 10\cos\left(2\pi(2000+\Delta f)t+\theta\right)$

$+10\cos\left(2\pi(18000+\Delta f)t+\theta\right) + 10\cos\left(2\pi(2000+\Delta f)t+\theta\right)$

i) $\Delta f = 100\,Hz \qquad \theta = 0$

$e(t) = 5\cos(2\pi 1100t) + 5\cos(2\pi 1100t) + 10\cos(2\pi 2100t) + 10\cos(2\pi 2100t)$

$= 10\cos(2200\pi t)$ Yorum: Mesaj işareti elde edilemez.

ii) $\Delta f = 0 \qquad \theta = \pi/2$

$e(t) = 10\cos\left(2\pi 1000t + \frac{\pi}{2}\right) + 20\cos\left(2\pi 2000t + \frac{\pi}{2}\right)$

$= -10\sin(2000\pi t) - 20\sin(4000\pi t)$

Yorum: Mesaj işaretine 90° faz farkı eklenmiştir.